

Public Economics

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Discussion of Homework Assignments II



Open Questions and Problems?

- A person's preference for consuming a bundle of goods x and leisure F is described by the utility function $U = x^{0.4} \times F^{0.6}$. The **price for a bundle of goods x** is **10€**. The person has a total of **60 hours** available. The **wage** for one hour of work is **30€**. The government levies a proportional income tax at a **tax rate of 25%**.
- a) Calculate the optimal distribution between work time and leisure for the person before the introduction of the tax. State the quantity of hours worked and the quantity of goods consumed.
- b) Calculate the optimal distribution between work time and leisure for the person after the introduction of the tax. State the quantity of hours worked and the quantity of goods consumed.
- c) How does the person react to the wage decrease due to the introduction of the tax? Does the substitution effect or the income effect dominate?

a) Calculate the optimal distribution between work time and leisure for the person before the introduction of the tax.

Hours Worked

1. Step: Set up budget constraint: $10x = (60 - F) * 30$

$$x = 180 - 3F$$

Relation between Wage/Price of Consumption

2. Step: Determine marginal rate of substitution: $MRS_{F,x} = -\frac{\frac{\partial U}{\partial F}}{\frac{\partial U}{\partial x}} = -\frac{0,6F^{-0,4}x^{0,4}}{0,4x^{-0,6}F^{0,6}}$

$$MRS_{F,x} = -1,5 \frac{x}{F}$$

We determine the marginal utilities and divide them by each other

3. Step: MRS = slope of budget constraint:

$$-1,5 \frac{x}{F} = -3$$

$$x = 2F$$

Tangent point -> Marginal utility of leisure relative to consumption corresponds to wage relative to price of consumption.

4. Step: Substitute x from 3 into the budget constraint:

$$2F = 180 - 3F$$

Labour hours: $60 - F = 24$

$$F = 36$$

$$x = 72$$

b) Calculate the optimal distribution between work time and leisure for the person after the introduction of the tax.

1. Step: Set up budget constraint with tax:

$$10x = (60 - F) * 30 * (1 - 0,25)$$

w is reduced by the tax.

$$x = 135 - 2,25F$$

2. Step: Determine marginal rate of substitution: It does not change due to the tax, as this does not affect the utility function!

$$MRS_{F,x} = -1,5 \frac{x}{F}$$

New relationship $w_{net}/p_x = -2,25$
(Price of leisure/Price of consumption)

3. Step: MRS = slope of budget constraint:

$$-1,5 \frac{x}{F} = -2,25$$

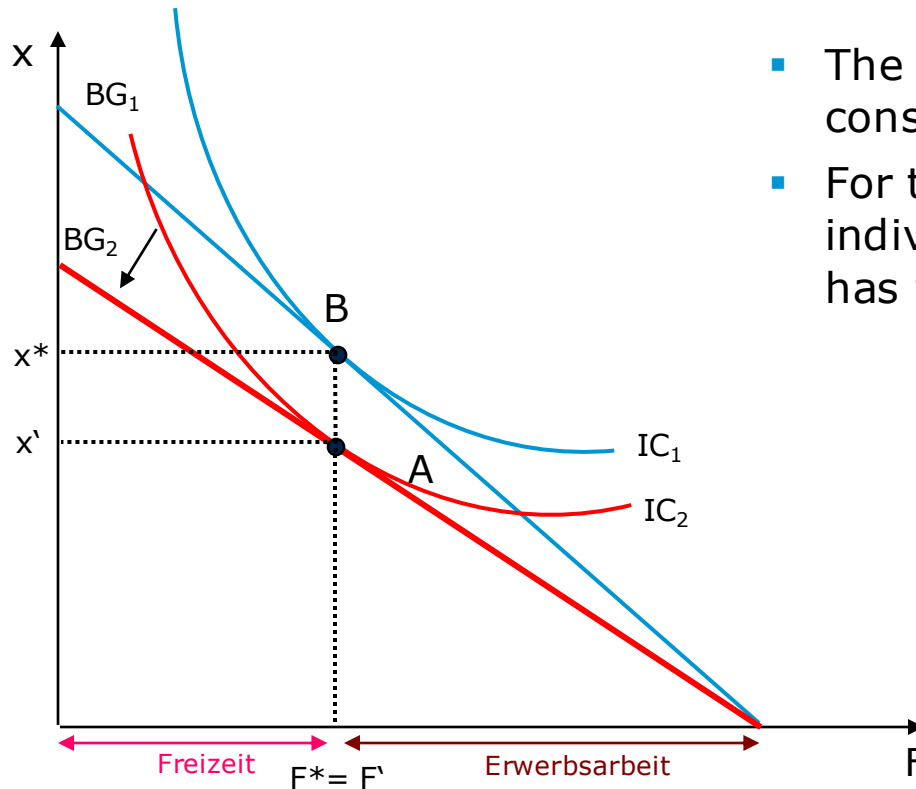
$$x = 1,5F$$

4. Step: Substitute x from 3 into the budget constraint:

$$1,5F = 135 - 2,25F$$

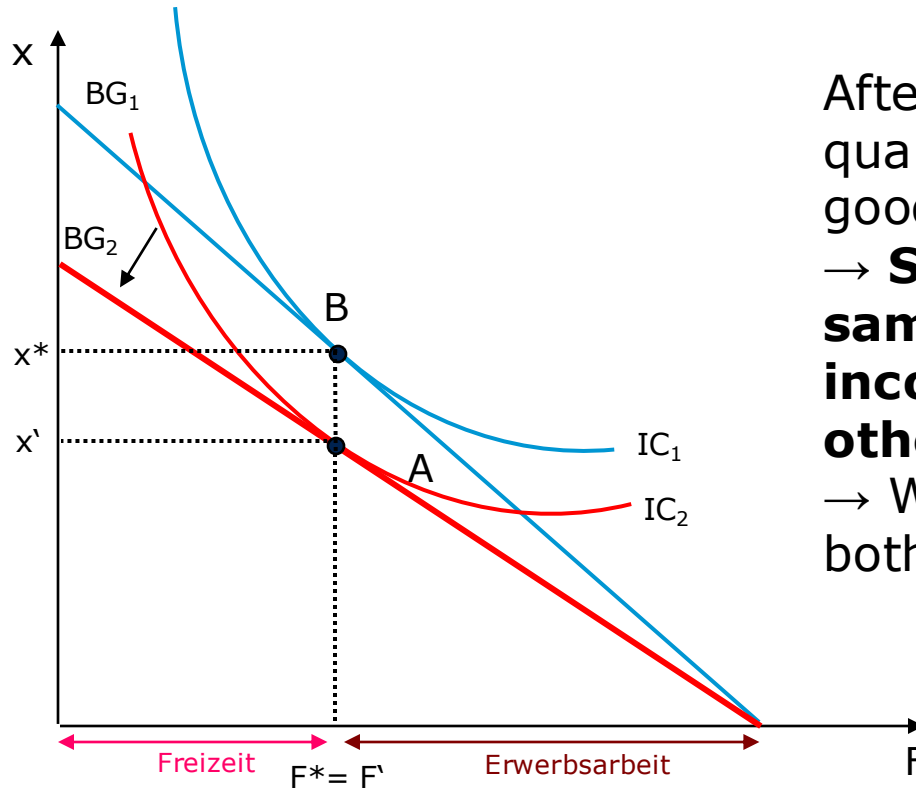
$$F = 36$$
$$x = 54$$

c) Describe how the person reacts to the wage decrease due to the introduction of the tax. Does the substitution or income effect dominate?



- The wage decreases → The budget constraint pivots inwards.
- For the same amount of work time, the individual receives less wage and thus has fewer consumption possibilities.

c) Describe how the person reacts to the wage decrease due to the introduction of the tax. Does the substitution or income effect dominate?

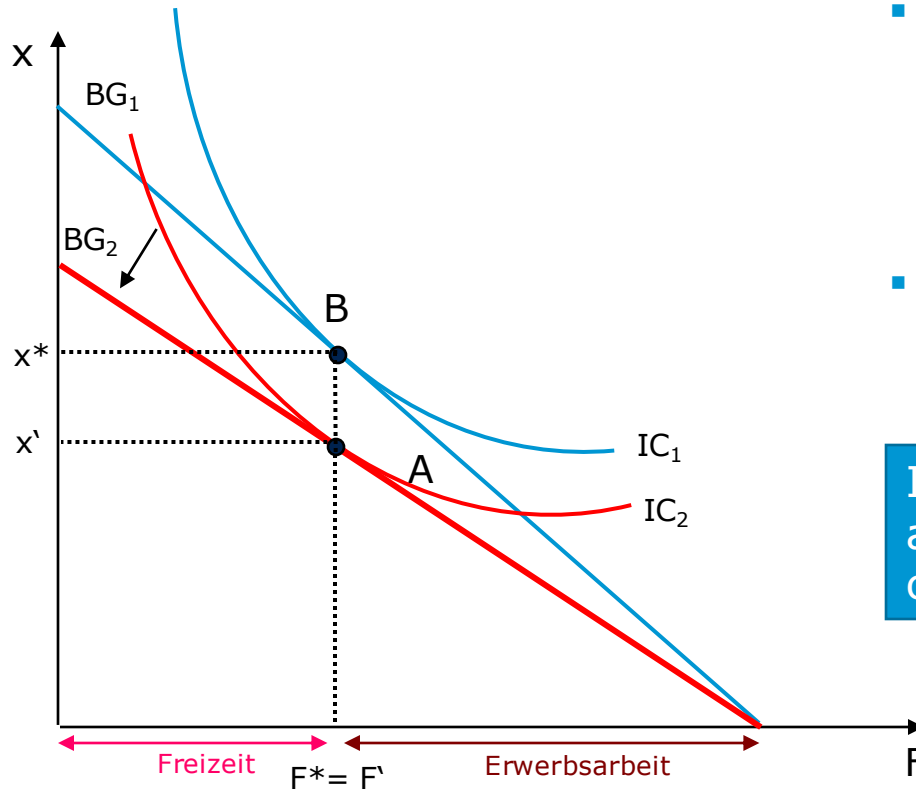


After the introduction of the tax, the quantity of consumed bundles of goods decreases.

→ **Since leisure remains the same, the substitution and income effects cancel each other out.**

→ Who can give me intuition on both effects?

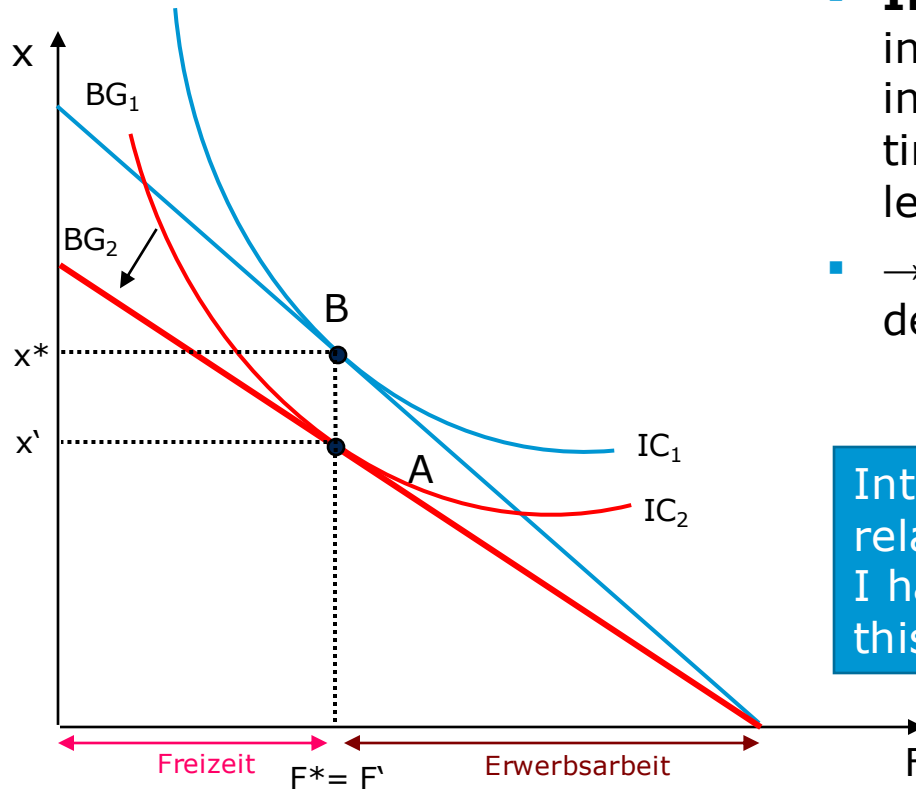
c) Describe how the person reacts to the wage decrease due to the introduction of the tax. Does the substitution or income effect dominate?



- **Substitution Effect:** Due to the introduction of the tax, the opportunity costs of leisure decrease → "it becomes cheaper to consume leisure."
- → Work time decreases & leisure increases

Intuition: The substitution effect always relates to the opportunity costs of leisure - do these rise/fall?

c) Describe how the person reacts to the wage decrease due to the introduction of the tax. Does the substitution or income effect dominate?



- **Income Effect:** Due to the increase in taxes, the individual receives less income for the same amount of work time. In order to maintain the income level, more work must be done.
- → Work time increases & leisure decreases

Intuition: The income effect always relates to a fictional income level - do I have to work more/less to maintain this?

Paper-Discussion

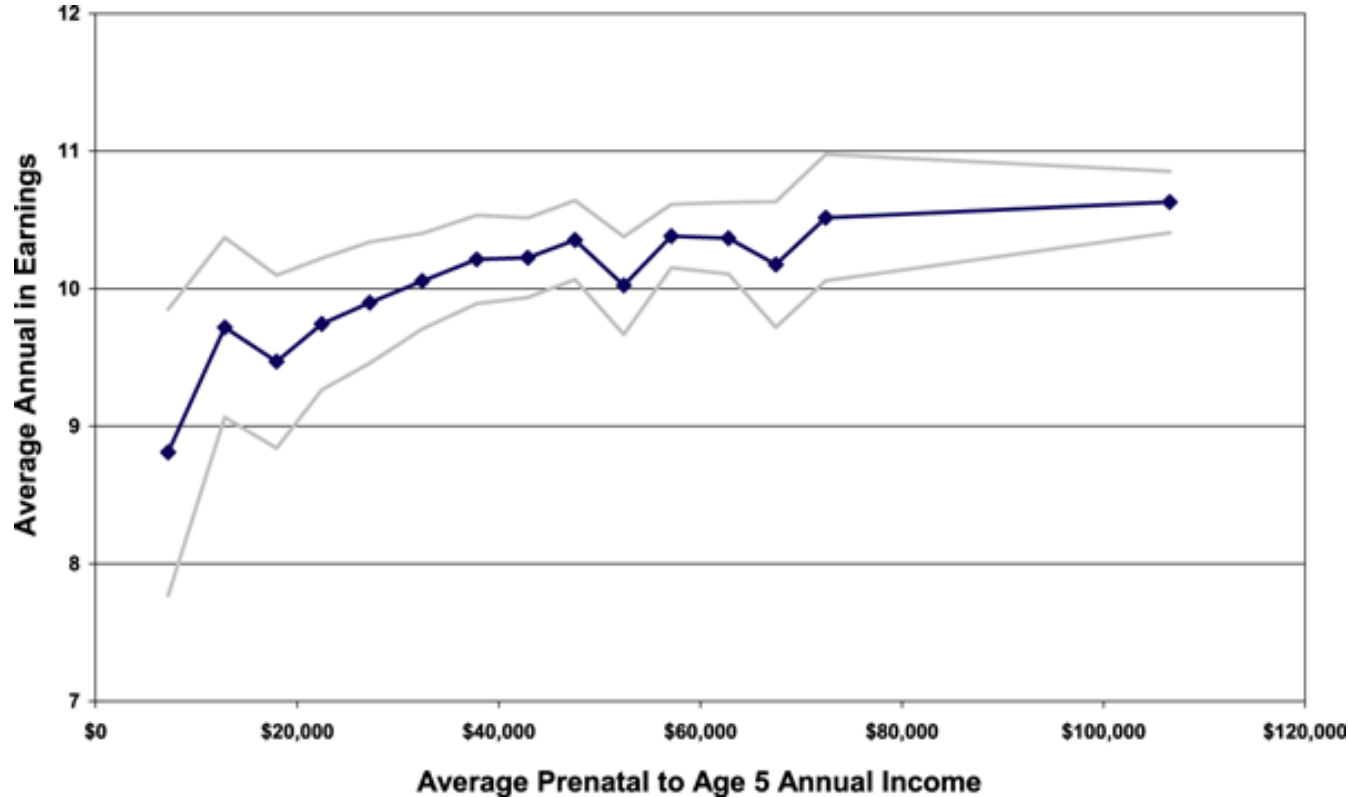


Barr, A., Eggleston, J., & Smith, A. A. (2022). Investing in infants: The lasting effects of cash transfers to new families. *The Quarterly Journal of Economics*, 137(4), 2539-2583.

We will discuss the paper using tables and figures in 4 groups:

- **Figure 1 & 2:** Briefly explains the RDD design in this study. Explains 2 specific assumptions that must be met.
- **What** is an "Intent-to-Treat (ITT)" parameter and why do the authors use it in their analysis? How can this be criticized?
- **Figure 3 & 4, Table 5:** What do the figures show? Why might the results for women differ? What does Table 5 show?
- **Figure 6:** What does the figure show? What hypothesis is this analysis intended to support? What can you criticize (also with reference to [Appendix Figure A.XIV](#))?

↑ Family Income => ↑ Children's Income



- **Influence**
- Upbringing?
- Place of residence?
- Neighborhood?
- Parents' education?
- ...
- Higher financial benefits?
- Causal?

"Parental Resources and College Attendance: Evidence from Lottery Wins" (Bulman et al., 2021)

- **Goal:** Investigation of the influence of financial resources on the college attendance of US children whose parents won the lottery.

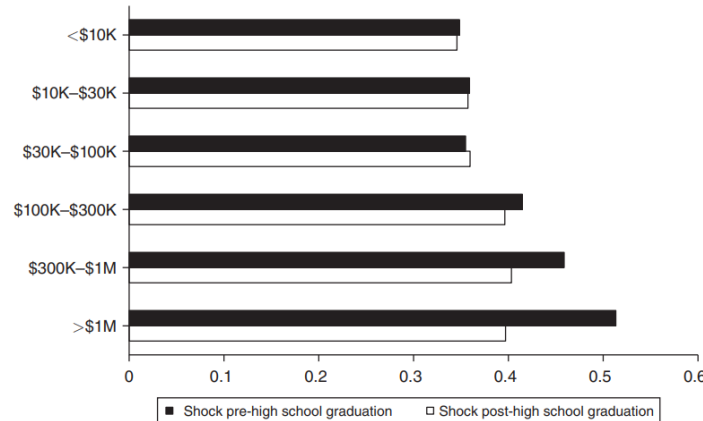


FIGURE 1. COLLEGE ATTENDANCE WITHIN ONE YEAR OF HIGH SCHOOL GRADUATION BY THE AMOUNT AND TIMING OF THE LOTTERY WIN

Notes: This figure presents the average rate of attending any college for children who graduate before and after their parent wins a lottery. Attendance is measured in the year of expected high school graduation and rates are adjusted for cohort fixed effects. Win sizes are adjusted to 2010 dollars and are classified according to five cutoffs: \$10,000, \$30,000, \$100,000, \$300,000, and \$1,000,000.

- Very weak effect per dollar.
- Higher effects only with large winnings that are well above the cost of tuition.
- Smaller effects for households with low socioeconomic status (SES).

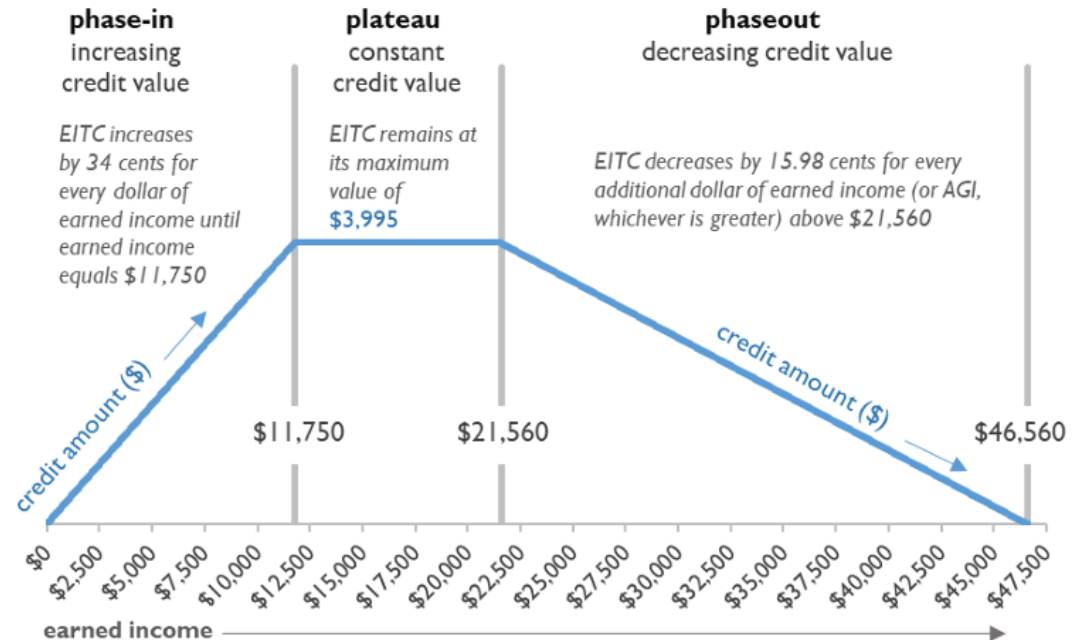
The Impact of the EITC on Children's Long-Term Outcomes

- **Several channels of influence of the EITC (extensive, intensive, SE, EE):**
- **Extensive Margin:** Work is more attractive than non-work.
- **Intensive Margin:**
- **Substitution Effect:** Mothers could increase their working hours in response to the EITC. This could mean more work time and less time with children.
- **Income Effect:** The EITC offers additional resources to low-income households, which could lead to less work time and more time with children.
- **Total Effect:** The total effect of the EITC on children's long-term outcomes is unclear.
- **Previous Findings:** The EITC leads to increases in household income (mothers work more), with only slight decreases in the time mothers spend with their children (Bastian, J., & Michelmore, K., 2018).

Incentive Phases of the EITC

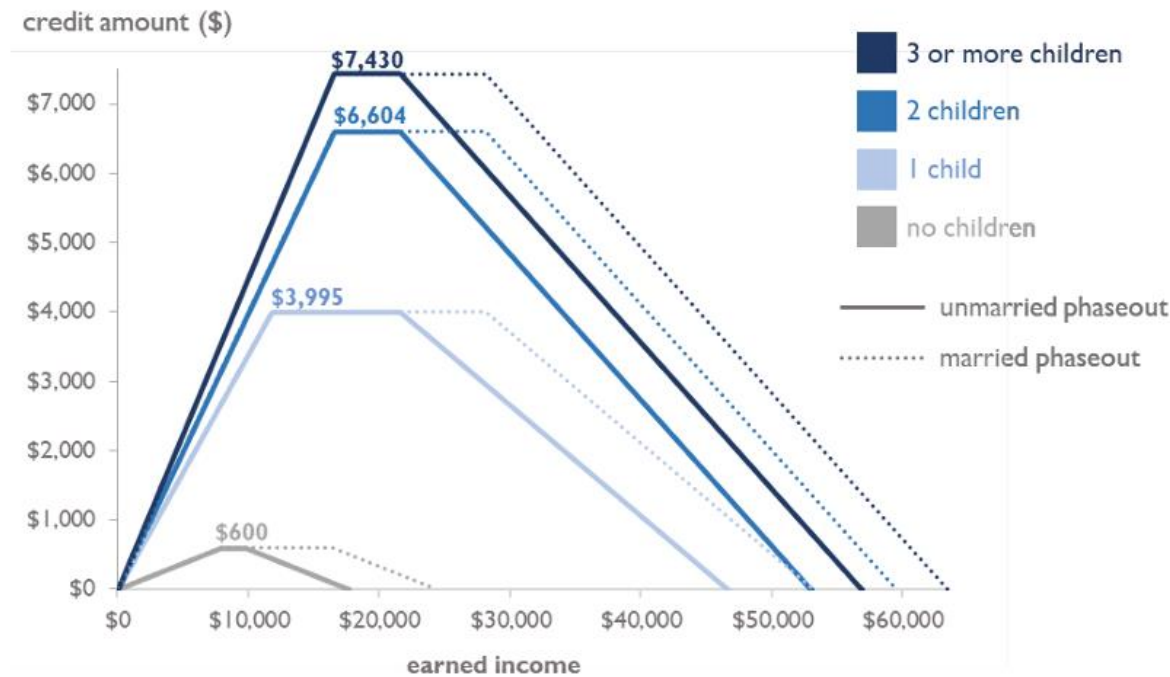
- *[I]n the phase-in region, the EITC leads to an ambiguous impact on hours worked due to the negative income effect and positive substitution effect.*
- *In the flat region, however, the EITC produces a negative income effect leading to an unambiguous reduction in hours worked.*
- *In the phase-out region, the EITC produces a negative income and negative substitution effect leading again to an unambiguous reduction in hours worked.*

Figure 2. EITC for an Unmarried Taxpayer with One Child by Income, 2023



EITC after kids

EITC Amount by Number of Qualifying Children, Marital Status, and Income, 2023



Source: CRS calculations based on IRS Revenue Procedure 2022-38 and Internal Revenue Code §32.

- **Register Data**
- **Taxes** (IRS 1040: Income Tax Return)
- **Social Security Data:** Gender, Date of Birth,..
- .Date of Birth => Financial benefit through EITC
- January born (+-14 days) for 1981-82, 1985-86, 1991-92

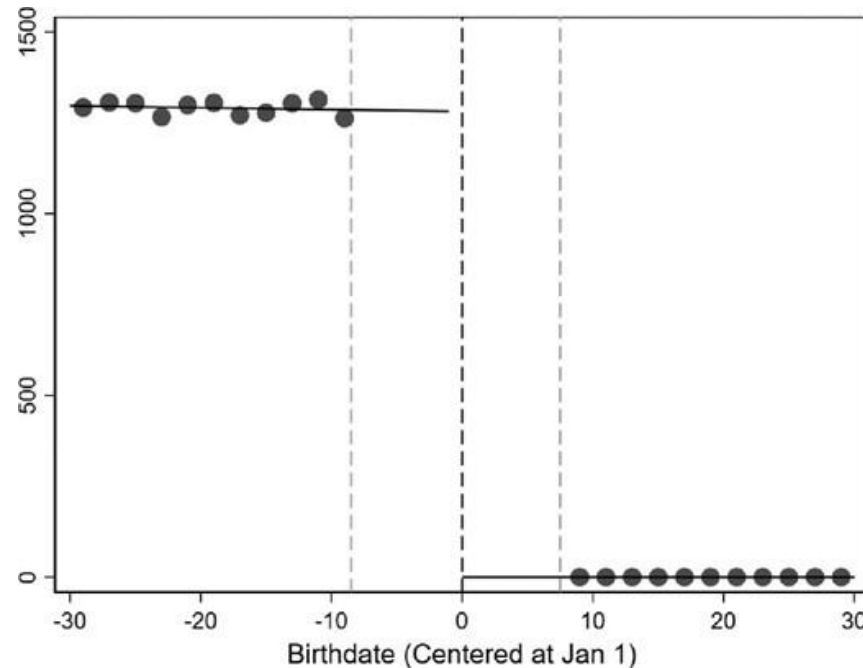
- **Linked with Parents, Parents' income** calculated shortly before birth.
- "predicted AGI" (Income) based on the previous year to reduce endogeneity
- Only for families who are EITC eligible (income threshold)

- **Register Data Education** ("North Carolina Education Data")
 - Problem?
 - Not linked with other register data!
 - But, exact birth dates of students.
 - **Proxy for EITC** = Free or Reduced-Price Lunch (FRL)
 - “For example, for a family of three with one child in 2000, the income cutoffs for eligibility were \$25,600 (FRL) and \$27,400 (EITC).”

- **Figure 1 & 2:** Briefly explain the RDD design in this study. Explain 2 specific assumptions that must be met.

Regression-Discontinuity-Design

- Children before 31.12 and after 01.01: Additional EITC benefits. Why?



- 31.12 => Full year of EITC benefits => **USD 1300 more!**

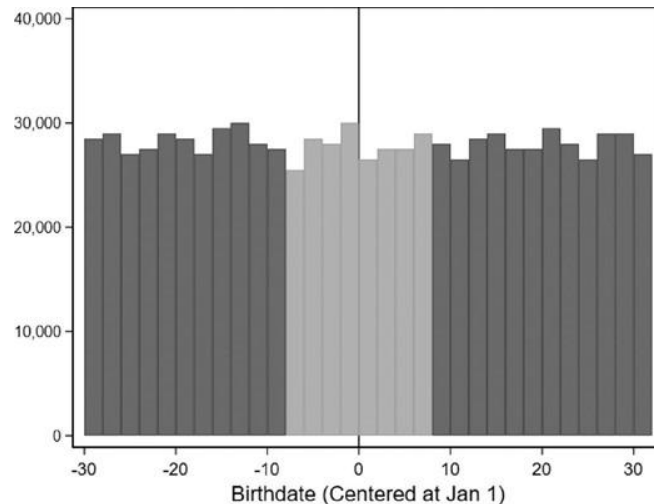
Regression-Discontinuity-Design

$$Y_{it} = \beta_0 + \beta_1 1[z_i < 0] + \beta_2 z_i + \beta_3 1[z_i < 0] \times z_i + \theta_t + \epsilon_{it},$$

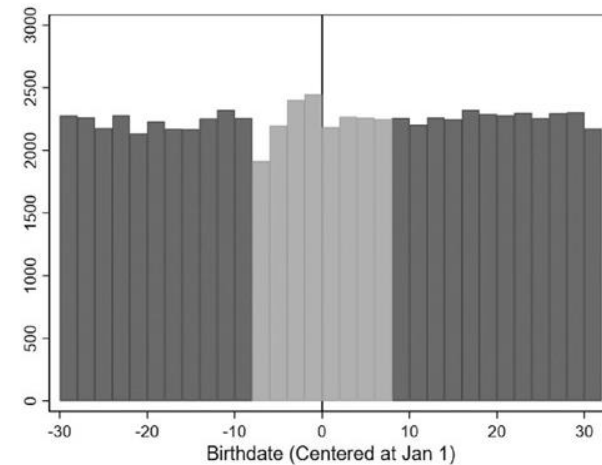
- **Y_{it}**: Income or education
- **Z_i**: is the difference between the child's date of birth and January 1st. $1[z_i < 0]$ is an eligibility indicator
- **θ_t**: Fixed effect of the year of birth (temporal trends, cohort effects)
- **Primary Coefficient (β₁)**: Measures the effect of the expected eligibility for child-related tax benefits for low-income families.
- **Secondary Coefficient (β₂)**: The effect of the actual date of birth (z_i) on the considered outcome (Y_{it}), such as school performance or the child's income.

RD Assumptions: Treatment Random

- **Treatment Assignment z_t is random!**
- Earlier birth to receive benefits?
- Only minimal effects... Robustness: Donut-Hole (without Light Gray)



(A) Tax Data (EITC-Eligible Families)



(B) North Carolina Data (FRL-Eligible Students)

RD Assumptions: Treatment Random

- **"Confounder"** of the Treatment
- **Children born after 31.12 start school later?**
- Federal states that have school start based on age within the January 1st window (light gray) were removed = 8% fewer.
- Further **Robustness Checks** (Differential analysis for parents with different cash benefits - apart from EITC)

RD Assumptions: Balancing

- Beta_1 as dependent variable =>
- No significant differences!

Balance on Baseline Characteristics

	Child	Child	Child	Child	Parent	Parent	Parent	Parent	Parent
	male	white	black	Hispanic	max. age	filed 1040	married	pred. AGI	in poverty
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Born before Jan 1	0.005	0.005	-0.003	-0.00	0.04	0.002	-0.003	19.35	0.001
	(0.004)	(0.006)	(0.003)	(0.005)	(0.07)	(0.003)	(0.003)	(71.67)	(0.003)
Mean	0.501	0.630	0.134	0.174	24.06	0.364	0.048	12,530	0.788

- **What** is an "Intent-to-Treat (ITT)" parameter and why do the authors use it in their analysis? How can it be criticized?

"ITT analysis includes every subject who is randomized according to randomized treatment assignment. It ignores noncompliance, protocol deviations, withdrawal, and anything that happens after randomization."

- **Primary Coefficient (β_1):** Anticipated eligibility for child-related tax benefits for low-income families.
- **Not whether tax benefits were actually received or not!**
- "Predicted" gross income based on the year before.
- Why?
- **To avoid "selection bias" and "endogeneity problems".**
- Adjust income in the birth year (certain years are not used, where there was a great tax incentive to report more/less).
- Avoid distortions due to different participation rates (selection = only the people who also benefit from the EITC or know how to apply for them).

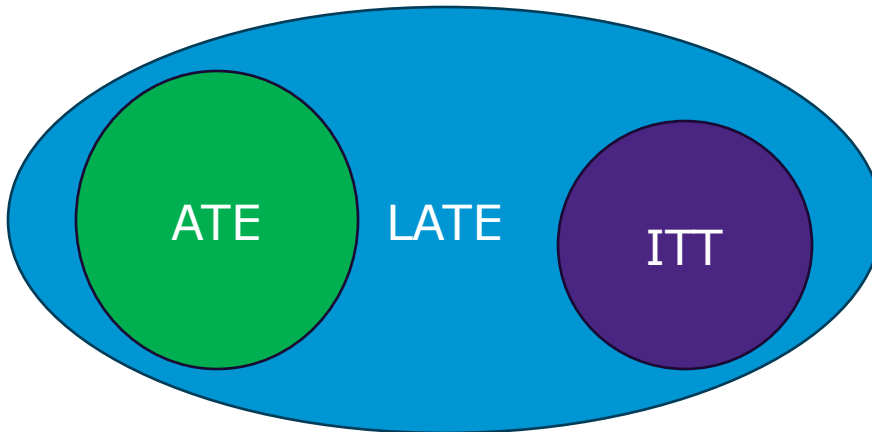
ITT, LATE, ATE

	Assigned to treatment	Assigned to control
Compliers	Treated	Not treated
Always-Takers	Treated	Treated
Never-Takers	Not treated	Not treated
Defiers	Not treated	Treated

- **ITT (Intent to Treat Effect):** Effect of the assignment to treatment (*not the actual participation*) on the outcome for everyone in the target group.
- Easy to calculate and robust against selection, but not the effect of the treatment!
- **LATE (Local Average Treatment Effect):** Effect of the treatment for the "compliers" – meaning those who follow the assignment. Often estimated by instrumental variables..
- **ATE (Average Treatment Effect):** Effect of the treatment on the outcome for all **individuals** in the population. Easiest to calculate if there is no "non-compliance" (in which case $ATE = LATE$).

Which Effect is Largest?

- Mostly: **LATE > ATE**
 - „Complier” effects are stronger, usually those who benefit the most from the treatment



ATE > LATE

Non-complier effects are stronger=> **very rare**

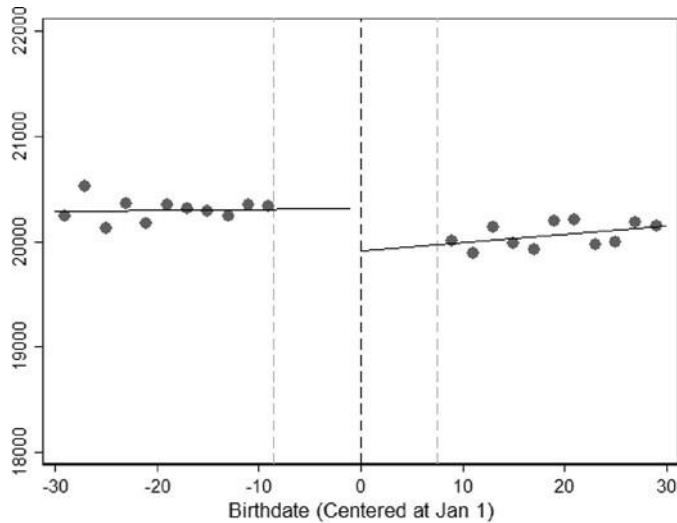
ATE = LATE:

Treatment effects are the same for all groups.
=> **also rare**

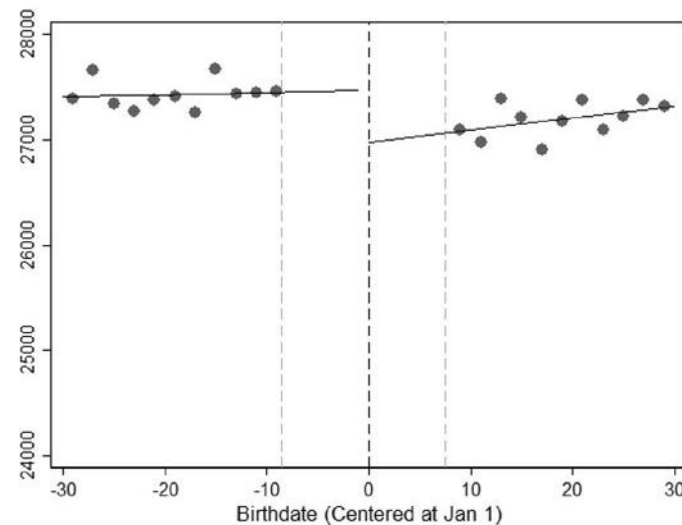
- **Graphs 3 & 4, Table 5:** What do the graphs show? Why could the results for women differ? What does Table 5 show?

Results: Income of the Children

- **Difference:** USD 319 (+1.6%) USD 456 (+1.7%)



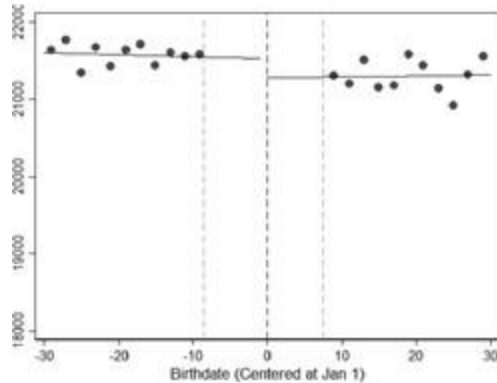
(A) Earnings (23 to 25)



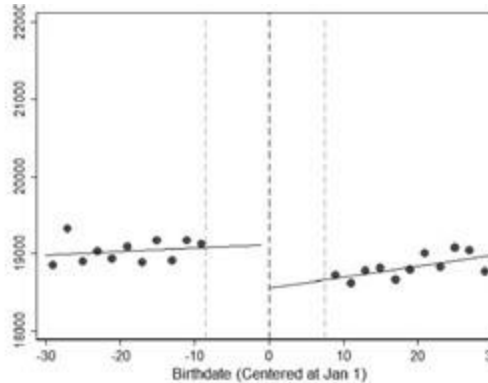
(B) Earnings (26 to 28)

- **Average additional benefits for families:** USD 1.291

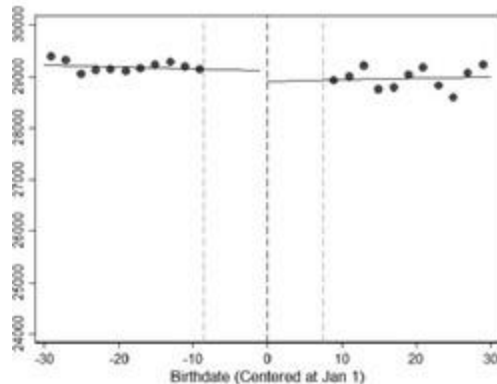
Household Level:



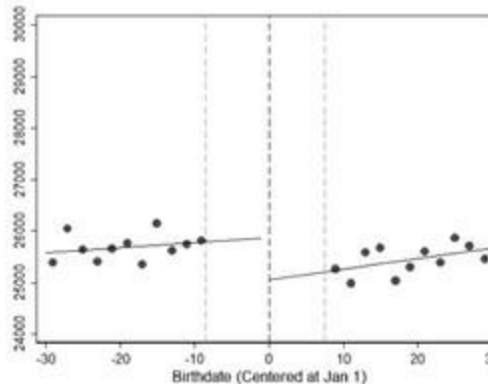
(A) Female Earnings (23 to 25)



(B) Male Earnings (23 to 25)



(C) Female Earnings (26 to 28)



(D) Male Earnings (26 to 28)

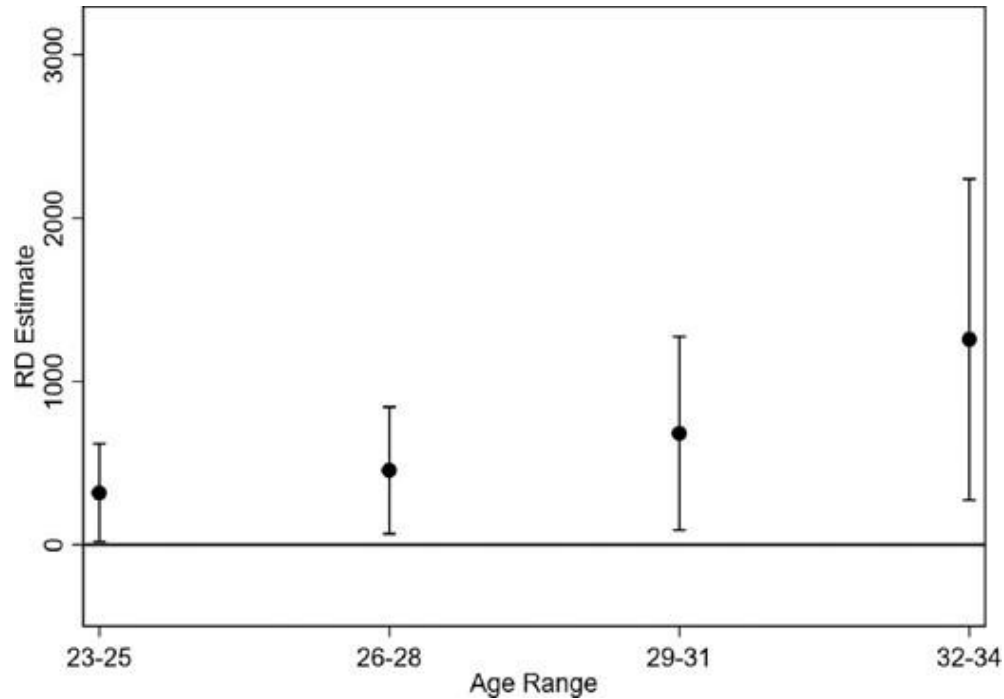
- **Smaller effects for Women:**
- Income at the household level
 - Stronger effects for Men, because
 - Women who marry => Smaller share of household income
- **Or do Boys benefit more from increased family income?**
- **Weaker evidence for this in this paper!**

Race/Ethnicity

- Strongest effects for Men
- Low effects for Hispanics
- why?
 - Fewer EITC applications from Hispanics! (Information asymmetry)

	Black			Hispanic			White (non-Hisp)		
	All	Male	1991-92	All	Male	1991-92	All	Male	1991-92
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Earnings (23-25)	944.0 ^{**}	1,075.0 [*]	1,836.0 ^{**}	614.6	67.8	276.0	55.3	628.7 [*]	624.0 [*]
	(472.7)	(601.0)	(753.2)	(521.0)	(582.2)	(691.2)	(201.8)	(337.5)	(327.7)
Mean	21,460	20,220	20,970	21,280	20,200	20,500	21,570	20,450	20,910
Earnings (26-28)	1,237.0	883.3	1,644.0	617.0	-394.4	120.0	519.8	1,370.0 ^{***}	1,048.0 ^{**}
	(810.5)	(1,048.0)	(1,266.0)	(727.8)	(1,005.0)	(978.4)	(325.6)	(483.2)	(510.0)
Mean	29,310	27,590	28,260	29,270	27,700	28,310	29,300	27,690	28,170
Cash transfer in infancy	1,306	1,306	1,808	1,317	1,317	1,808	1,313	1,313	1,808
Observations	66,000	32,500	26,000	87,000	43,000	34,500	313,000	153,000	124,000

Aggregated Income

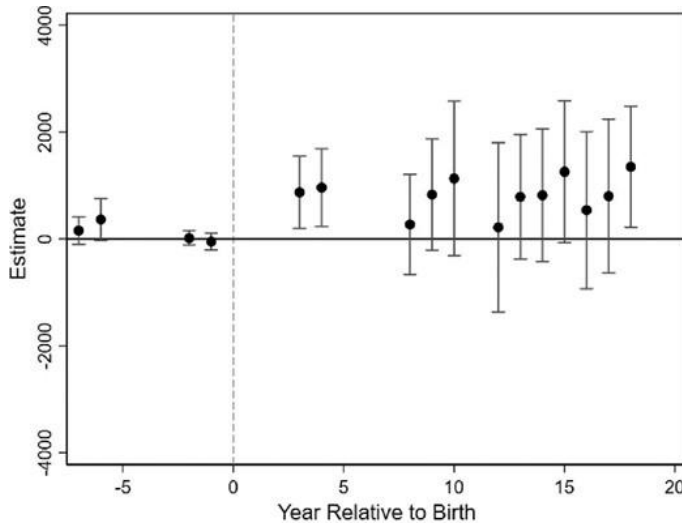


- Sustained and even increasing income effects.

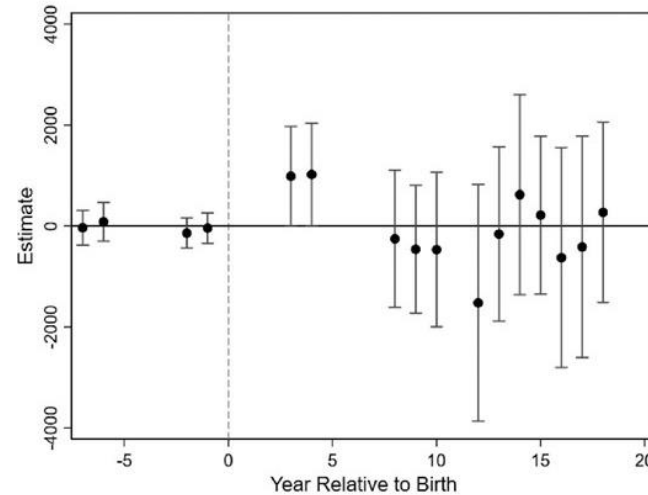
- **Graph 6:** What does the graph show? Which hypothesis is this analysis meant to support? How can you criticize it (also refer to [Appendix Figure A.XIV](#))?

Mechanisms Behind the Effects

- **Cash benefits** between the Control/Treatment Group in aggregate (over 18 years) are **relatively similar** or could even be the same!
- *Hypothesis:*
- **H: Liquidity Boost at an important time for raising children!**
- => Family income increases:



(A) Full Sample

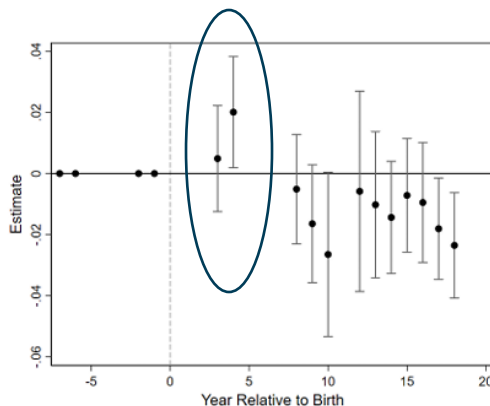


(B) Filers

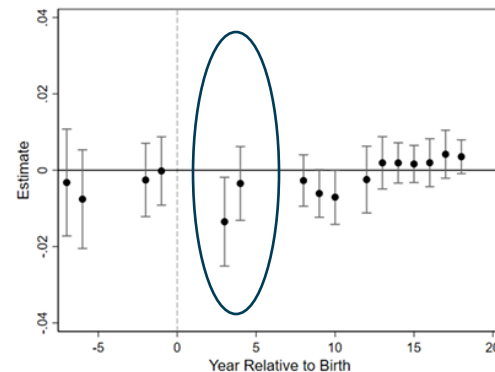
- Long-term unclear for "Filers" => better income data needed, as a regular explanation is made. Important to look closely! All families vs. "Filers"

Influence on Children, Poverty, Marriage, and Income Tax

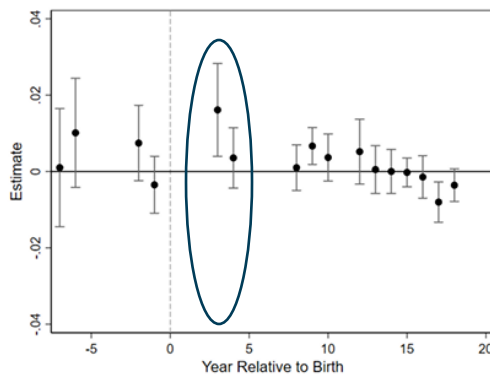
- **Dependents" = Children**
- More children, but?
- **1040: Income Tax**
- Proxy for work.
- External Margin



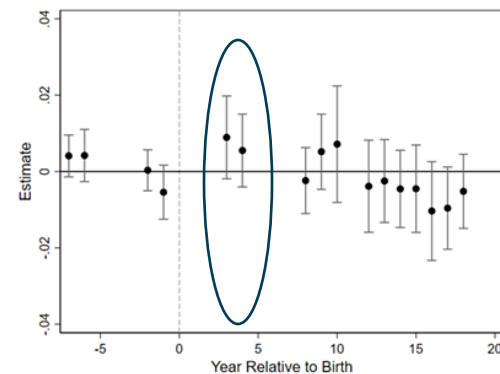
(a) Dependents



(b) Parent Poverty Status



(c) Parent File 1040



(d) Parent Marital Status

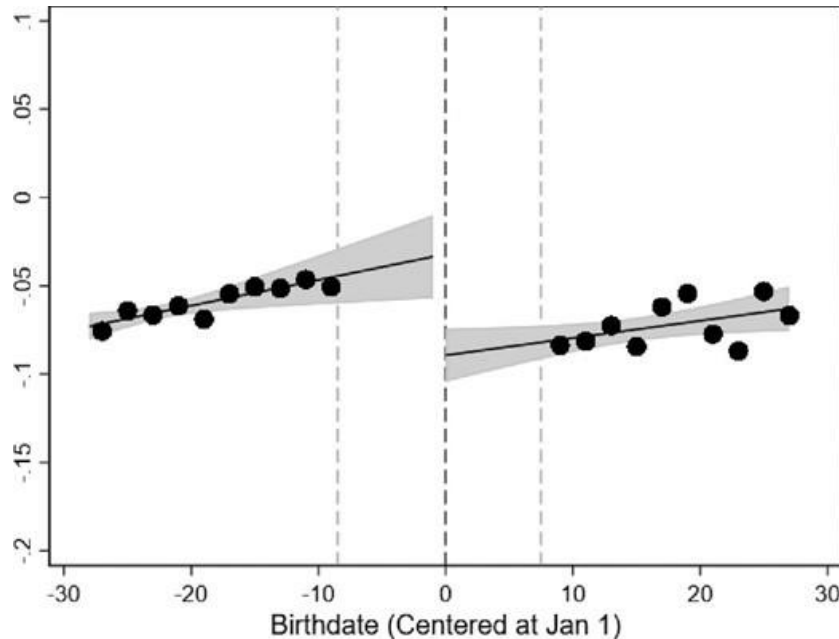
In Aggregate: Not quite so robust...

- "Possible reduction of poverty status"
- Only at certain ages
- Therefore, Event-Study Plots!
- **"Tendency for fewer divorces"**
- Married and Single Estimates
- => **"Tendency for partners"**
 - „In combination, the estimates suggest that the resource boost following the birth of a first child may have had positive and enduring effects on the parents“

Table A.IX: Effect of Cash Transfer Eligibility on Parent Outcomes

	Control Mean (1)	RD Estimate (2)
Cumulative Earnings	625,600	11550** (4903)
Cumulative Earnings (Filer) ^a	737,200	7493 (7103)
Mean Earnings	58,250	903.7** (458.7)
Mean Poverty	0.21	-0.001 (0.002)
Always in Poverty	0.04	0.001 (0.002)
Always Filer	0.46	0.007** (0.003)
Ever Married	0.71	-0.002 (0.005)
Ever Single	0.84	-0.005* (0.003)

Influence on Education



- Eligibility for free meals as a discontinuity.
- Index (Educational attainment, behavior, aptitude/talent)
- **0.05 standard deviation higher index!**
- **=> 11% difference**
- **No difference between genders** (maybe due to measurement inaccuracies in household income)
- Similar effects for Math/German tests
- 1.4% graduation and 1.4% less likely to "drop out"

Summary and Implications

- **Long-term effects:** Positive effects on income and educational outcomes are still visible in adulthood.
- USD 1000 per family => **1%-2% higher income in 20s/30s!**
- **11% higher education index.**
- **Policy Implications:** Evidence that early financial transfers **represent a sustainable investment in social mobility.**
- **Cost-benefit:** The **increased tax revenues** in adulthood could **offset** or even **exceed the initial transfer costs.**

- More detailed analysis of the incentive dynamics of **Transfer payments**
- Theory
- Example: Basic income
- Example: Unemployment insurance
- Special case: Information asymmetry